

1. Project Title: AI-Based Disaster Response Management System for Resource Allocation and Relief Coordination

2. Project Statement and Outcomes:

During disaster events, resource allocation decisions for food, water, medical supplies, search and rescue teams, and shelter are often made with incomplete information, resulting in critical shortages in some areas while resources sit idle elsewhere. This project builds a full-stack AI-powered disaster resource management web platform that integrates real-time disaster event feeds, geospatial damage assessment data, population density maps, and field resource inventory to model disaster impact zones, predict resource demand by location, and generate optimized allocation and logistics plans for relief agencies.

The deliverable is a disaster operations command platform where relief coordinators manage their resource inventory, view real-time disaster impact maps, and receive AI-generated resource allocation recommendations with deployment routes. Field teams access mobile-friendly mission assignments with supply manifests and navigation support. Post-event analytics modules generate comprehensive situation reports

supporting debriefing, donor accountability, and future preparedness planning.

3. Modules to be Implemented:

1. Disaster Data Ingestion & Geospatial Impact Assessment
2. Resource Demand Prediction Engine
3. Logistics Planning & Disaster Operations Dashboard
4. System Integration, Testing & Project Finalization

4. Week-wise Module Implementation and High-Level Requirements:

Weeks 1–2: Disaster Data Ingestion & Geospatial Impact Assessment

- Integrate real-time disaster event data feeds from GDACS, USGS (earthquakes), NDMA, and satellite imagery APIs for damage extent estimation.
- Build a geospatial impact assessment module combining flood/earthquake extent maps, population density rasters, and building footprint data (OpenStreetMap).
- Develop a resource inventory management system for tracking available quantities of food packets, water, medical kits, rescue vehicles, and personnel.

- Build a geographic zone classification system dividing disaster-affected areas into grid cells with impact severity scores.
- Set up a PostGIS-enabled geospatial database for storing disaster zones, resource depot locations, and field operation records.

Weeks 3–4: Resource Demand Prediction Engine

- Train a regression model predicting resource demand (food packets, water, medical kits, shelter capacity) per affected zone using historical disaster response data.
- Build a vulnerability weighting module adjusting demand estimates for zones with higher proportions of elderly, children, or medically dependent populations.
- Implement a dynamic demand update engine that recalibrates predictions as new field reports and satellite assessments arrive.
- Build scenario simulation functionality allowing relief coordinators to model different response speeds and resource availability scenarios.
- Expose the demand prediction service via a FastAPI endpoint returning zone-level resource demand estimates with confidence intervals.

Weeks 5–6: Logistics Planning & Disaster Operations Dashboard

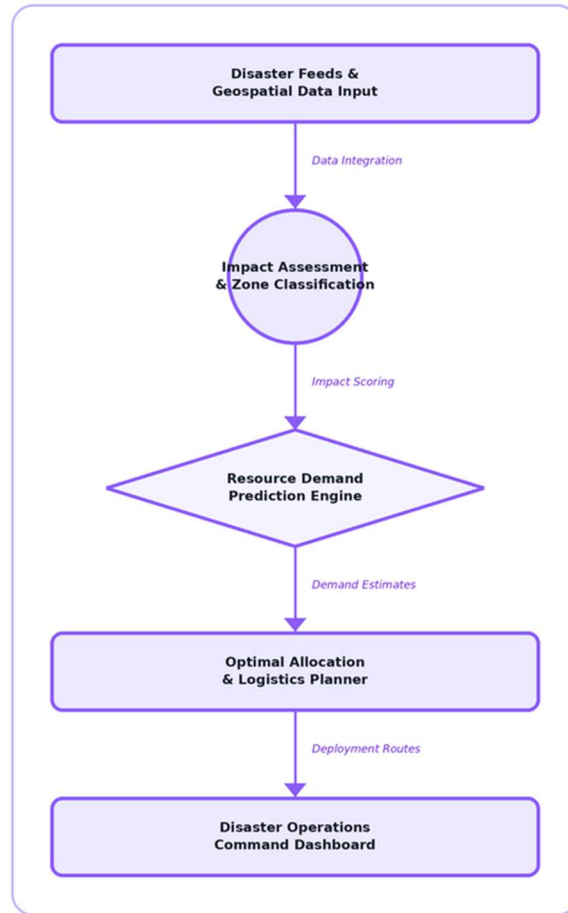
- Implement a linear programming / OR-Tools optimization model allocating available resources across demand zones to maximize coverage while respecting transport capacity and route constraints.
- Build a logistics routing engine generating optimized delivery routes from resource depots to affected zones, accounting for road accessibility post-disaster.
- Develop a priority override mechanism allowing commanders to manually elevate certain zone priorities in the allocation model.
- Develop a React.js / Leaflet.js command dashboard with a real-time disaster impact map, resource allocation overlays, and field team tracking.
- Build a mission assignment panel, resource tracking module, and a post-event analytics module generating situation reports; deploy on a high-availability cloud environment.

Weeks 7–8: System Integration, Testing & Project Finalization

- Conduct end-to-end integration testing across all modules — disaster feeds → impact assessor → demand predictor → logistics optimizer → command dashboard — and resolve any cross-module compatibility issues and edge-case failures.

- Perform load and stress testing on the backend APIs and cross-browser UI validation; address performance bottlenecks and implement enhancements from peer and mentor review. Conduct scenario testing using historical disaster datasets (Kerala Floods 2018, Chennai Floods 2015).
- Refine the user experience based on testing feedback: improve interface clarity, add contextual help tooltips, and ensure full responsiveness on mobile and tablet devices.
- Prepare comprehensive project documentation covering system architecture, data flow diagrams, API reference, model or algorithm evaluation report, deployment guide, and known limitations.
- Develop the final project presentation (PPT) covering problem statement, proposed solution, architecture, key features, technology stack, results, challenges, and future scope — and rehearse the full end-to-end demo for a smooth, confident delivery.

5. Flow Chart:



6. Sample Dashboard:

ReliefGrid

Disaster Relief Resource Allocator

Dashboard

Incidents

Resources

Volunteers

Map

Settings

AP

Admin

Dashboard

Active Incidents

Resources

Volunteers

Relief Map

Logistics

Reports

Settings

Disaster Relief Dashboard

AI-optimized resource allocation for emergency response

Active Incidents

12

4 critical

Resources Deployed

2,840

across 9 zones

Volunteers Active

1,248

in field

People Assisted

48,200

this event

Resource Allocation by Category (Units Deployed)

8420

12000

1240

2840

3640

48

Food Pk...Water (...)

Med Kits Tarpauli...

BlanketsRescue ...

Incident Status

Live

Critical (life risk)

4

High (shelter need)

3

Medium (food/water)

3

Recovery phase

2

Active Incident Summary

Incident ID	Type	Location	Affected	Resources Sent	Priority	Status
INC-2026-04	Flood	Wayanad, Kerala	12,400	Food+Water+Med	Critical	Active
INC-2026-03	Cyclone	Puri, Odisha	18,200	Shelter+Food	Critical	Active
INC-2026-02	Earthquake	Manipur	8,400	Med+Rescue	High	Stabilising
INC-2026-01	Drought	Vidarbha, MH	9,200	Water+Food	Medium	Ongoing
INC-2025-18	Flood	Assam	0	All categories	Recovery	Closed